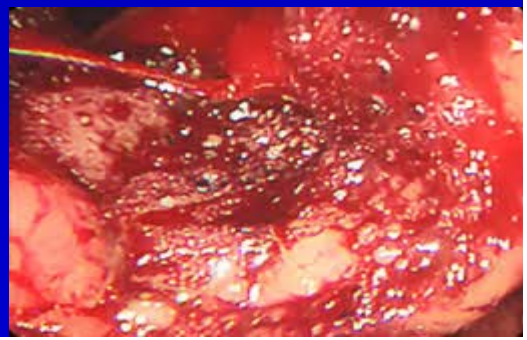
Treatment of Peptic Ulcer

Prof. Mona Kamal

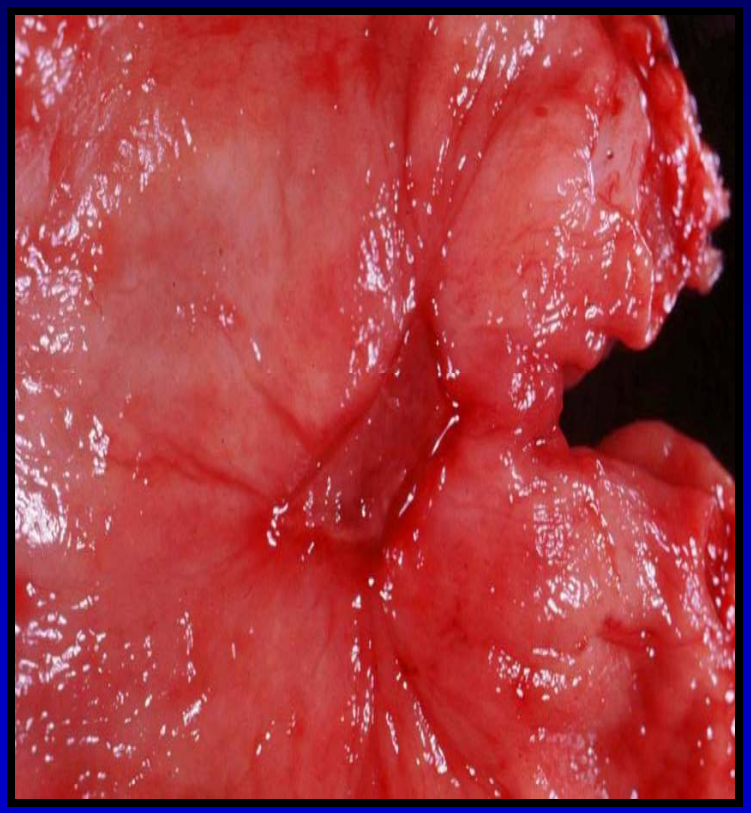
Objectives:

- Classify different drugs used in the treatment of peptic ulcer
- Explain the mechanisms of action of drugs used in the treatment of peptic ulcer
- Discuss adverse effects of drugs used in the treatment of peptic ulcer
- Discuss drug therapy for eradication of Helicobacter Pylori
- Design a treatment plan for patients with peptic ulcer disease



Peptic Ulcer

A break in the mucosa of the alimentary tract, which extends through the muscularis mucosa into the submucosa or deeper.



Gastric and Duodenal ulcers





Pathogenesis of Ulcers

Aggressive Factors

Acid, pepsin, gastrin
Stress
Smoking
Alcohol
Drugs (NSAIDs)
H. Pylori



Defensive Factors

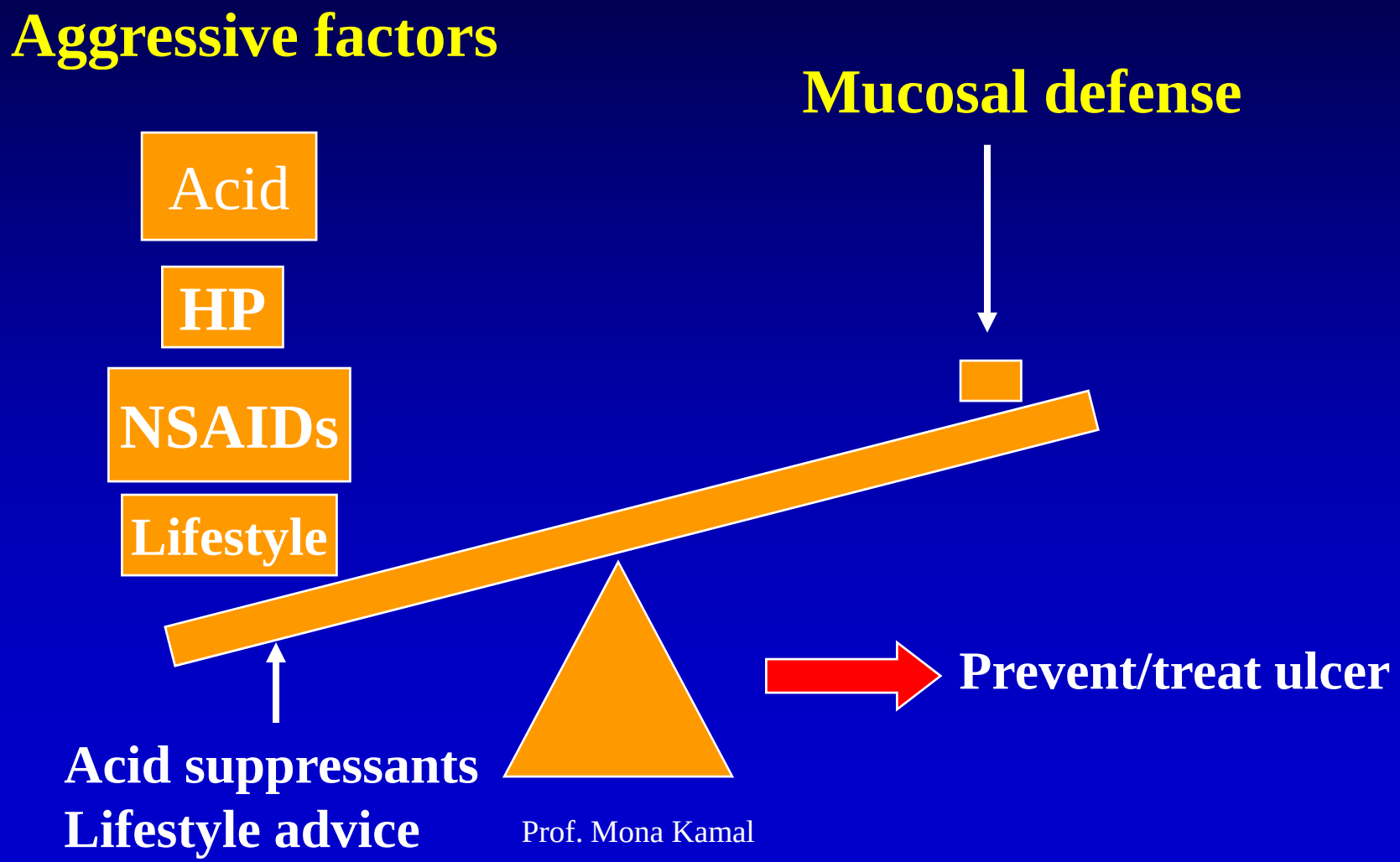
Mucus, bicarbonate layer
Blood flow, cell renewal
Prostaglandins
Free radical scavengers
Somatostatin

Helicobacter pylori

- **Gram negative bacilli**
- **Do not invade cells – only mucous**
- **Break down mucosal defense**
- **Chronic Superficial inflammation**



Therapy is directed at enhancing host defense or eliminating aggressive factors .



Aim of therapy

1. Relief pain
2. Promote healing
3. Prevent recurrence
4. Prevent complications

Lifestyle measures

- Stop smoking
- Avoid spicy foods , caffeine , alcohol
- Avoid Aspirin , other NSAIDs and corticosteroids
- Manage stress
- Small frequent meals

Drug therapy for peptic ulcer

- ↓ acid secretion

1. H_2 receptor antagonist
2. Proton pump inhibitors
3. Anticholinergic drugs
4. Gastrin antagonists
5. PGE-analogue

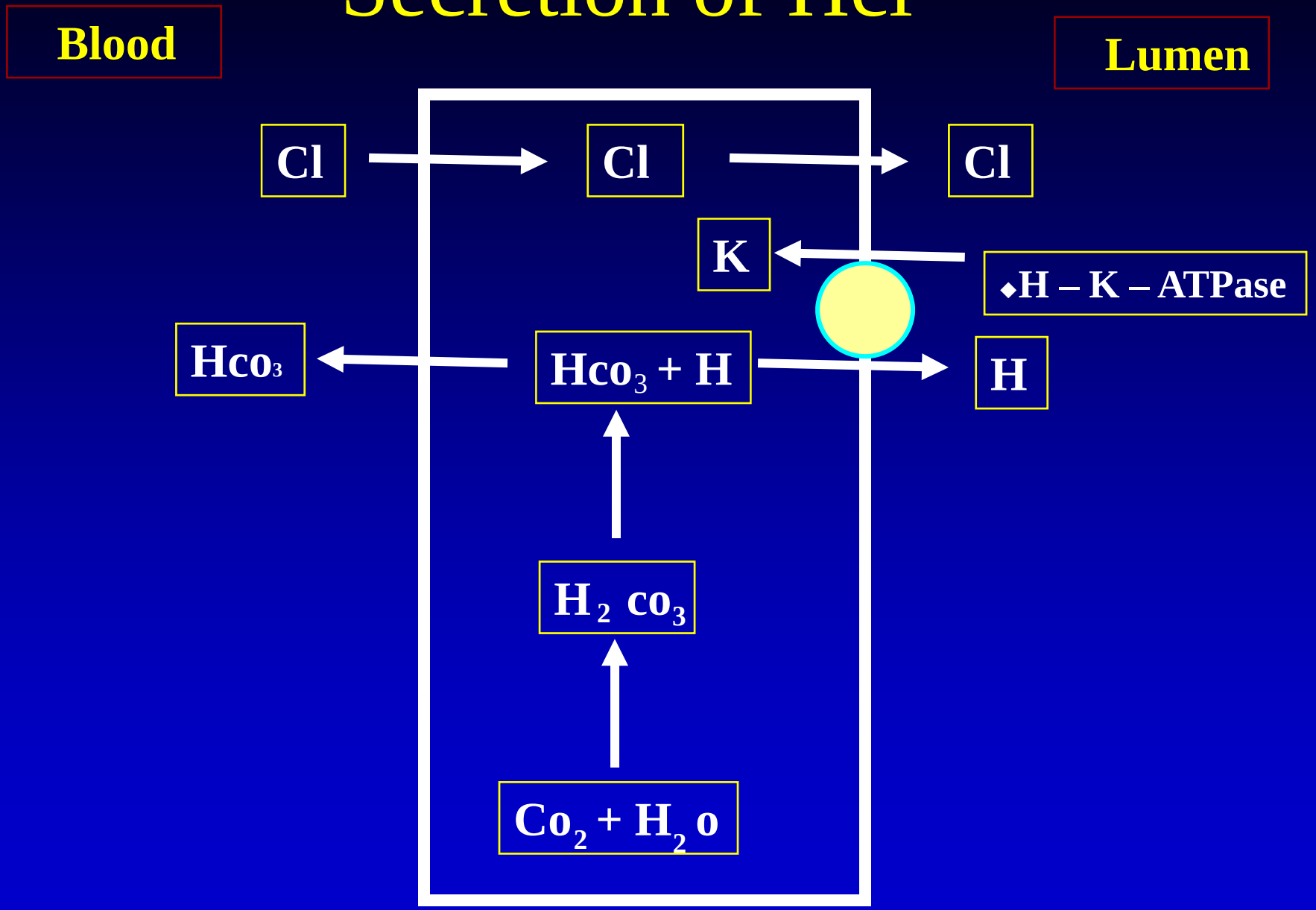
- Others

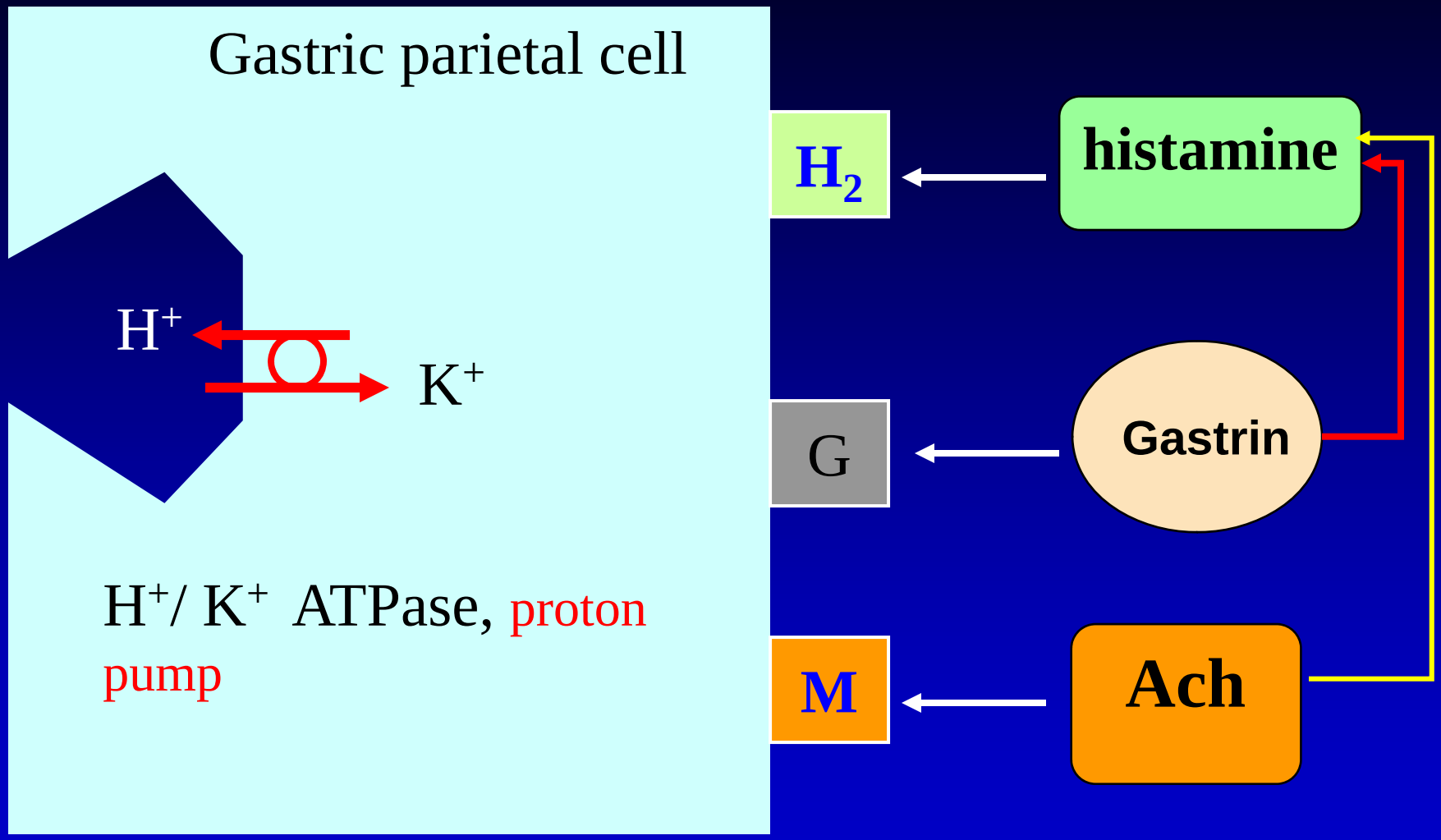
1. Acid neutralization: Antacids
2. Anti – H. pylori

- Mucosal protective factors:

1. Sucralfate.
2. Colloidal bismuth.
3. Prostaglandins (PGE)-analogue

Secretion of Hcl





Histamine is necessary for the action of gastrin & Ach

H_2 receptor antagonists

- Histamine is the predominant final mediators that stimulate parietal acid secretion.
- Act as competitive inhibitors of the H_2 receptors, resulting in decrease of all phases of gastric acid secretion " basal , nocturnal and stimulated". Thus abolishing HCl secretion mediated by histamine, while that mediated by Ach and gastrin is partially inhibited.
- Decrease formation of pepsin.
- Promote healing of duodenal & gastric ulcers

H_2 receptor antagonists Therapeutic uses

- potently inhibit (greater than 90%) basal, food-stimulated, and nocturnal secretion of gastric acid.
- 2nd line therapy for peptic ulcer &Zollinger-Ellison syndrome (Gasrtin-producing tumor of the pancreas)
- Prevention of stress ulcer in liver cirrhosis & anesthesia
- All H2 antagonists may reduce the efficacy of drugs that require an acidic environment for absorption, such as *ketoconazole.*

Used as single dose at bed time or twice daily for 4-8 weeks + eradication of H. pylori

Cimetidine

- Adverse effects:

1. Inhibitor of cytochrome p450, increase levels of warfarin ,theophylline and phenytoin.
2. Decrease hepatic blood flow , so decrease propranolol metabolism
3. Headache and confusion in the elderly as it pass the B.B.B
4. Cause aplastic anemia and thrombocytopenia
5. In males :Act as androgen receptor antagonist causing reversible gynecomastia, sexual dysfunction
6. In females :Increase secretion of prolactin with development of galactorrhea and infertility

Others

- Ranitidine:

It is more potent than cimetidine , has longer duration, does not bind to the androgen receptors, no increase in prolactin , no enzyme inhibition , not pass B.B.B.

Is relatively safe in pregnancy

- Famotidine: **Twice as potent as ranitidine, with longer duration**
- Nizatidine: **As potent as ranitidine**

Caution:

- Cimetidine, Ranitidine, Famotidine are metabolized in liver. Given carefully in liver disease.
- Nizatidine is mainly excreted by kidneys .

H_2 receptor antagonists

Drug interactions:

- Cimetidine inhibits liver microsomal enzymes (P-450) and can ↓ metabolism & ↑ toxicity of many drugs like: warfarin, theophylline, phenytoin, diazepam.
- Antacids can bind H_2 antagonists in the lumen of GIT & reduce their absorption
- H_2 antagonists reduce activation of PPIs

Combination of H2 antagonists with antacids or PPIs is not advised ?

Proton-Pump Inhibitors

Omeprazole , Lansoprazole Esomeprazole, Pentoprazole, Rabeprazole :

- They are weak bases enteric coated (destroyed by gastric acidity) from which they are readily absorbed in alkaline medium of the intestine as they are weak bases.
- They are prodrugs , after being absorbed in the intestine, they reach the gastric parietal cells through circulation to be activated in the acidic media to produce its effect.
- Causes irreversible inhibition of the parietal cell proton-pump (H^+/K^+ ATP-ase), blocking the transport of hydrogen into the lumen ; reducing both the basal and stimulated acid secretion.
- PPIs are the most effective acid suppressants as they act on the final step of HCl secretion. Almost 100 % control of HCl secretion

Proton-Pump Inhibitors

Omeprazole , Lansoprazole Esomeprazole, Pentoprazole, Rabeprazole :

- Rapid onset and long duration of action (action starts after 1 hr and last for 3 days.)
- ↓ pepsin activity
- Promote healing of peptic ulcer & prevent recurrence

Indications:

1. Drug of choice for Peptic ulcer
2. Drug of choice for Zollinger-Ellison syndrome

   Gasrtin-producing tumor of the pancreas
3. Ulcerative reflux oesophagitis
4. Prevention of stress- induced peptic ulcer in liver cirrhosis, anesthesia & surgery
5. Used with antimicrobial regimens to eradicate H. pylori.

Single dose daily for 4-8 weeks

Proton-Pump Inhibitors

Adverse effects:

1. Persistent low HCl may cause bacterial & fungal infections, diarrhea, Clostridium difficile colitis
2. CNS: Headache, dizziness
3. Deficiency of Vitamin B12 because acid is required for its absorption in a complex with intrinsic factor.
4. Osteoporosis and risk of fractures if the duration of use is 1 year or greater
5. Inhibit metabolism of warfarin, phenytoin & diazepam (microsomal enzyme inhibitor)
6. Prolonged inhibition of acid secretion (hypochlorhydria) : ↑ risk of gastric neoplasia , in rats **(safety?)**

Proton-Pump Inhibitors

Drug interactions:

- PPIs need activation in acid media, antacids & H_2 antagonists will ↓ this activation
- Antacids may also bind these drugs
- Liver enzyme inhibition by omeprazole decreases metabolism of warfarin, phenytoin & diazepam.

Combination of PPIs with antacids & H_2 antagonists is not advised?

Anticholinergic drugs

- Selective M_1 receptor antagonists: Pirenzepine, Telenzepine
- Inhibit gastric acid secretion with minimal unwanted effects of cholinergic blockade
- Less effective than other antisecretory drugs , so used as adjuvent with H_2 antagonists.

Gastrin antagonism

Gastrin : It is a peptide hormone released by G cells in the pyloric antrum of the stomach, duodenum, and the pancreas, then secreted into the portal blood , cause stimulation of acid secretion

(A) Proglumide: It is gastrin-receptor blocker

(B) Octreotide: It is long acting synthetic somatostatin analog , used parentally to decrease the release of gastrin

Protective agents: Misoprostol

- Congener of prostaglandin E_1 that acts on the parietal cells to:
  1. Stimulate bicarbonate secretion
  2. Stimulate mucus production.
  3. Increase blood flow
  4. Decrease Hcl secretion.
- Used in patients taking NSAIDs who are at risk for gastric ulcers (cytoprotective effect)
- May cause diarrhea and stimulation of uterine contraction . Avoided in pregnancy.

Protective agents: Sucralfate

- Sucrose sulphate+ Aluminum hydroxide gel
- Low pH in the stomach causes release of Alu leaving the compound with a strong negative charge , binding to the positively charged proteins in the ulcer base forming a protective barrier against acid, bile and pepsin. Only active in acid media .
- Forms a complex gel with mucus to prevent Hcl back diffusion , and prevent mucus degradation by pepsin.
- Stimulates mucus & bicarbonate secretion
- Increases PG secretion.

- Side effects: cause constipation , interferes with absorption of other drugs (digoxin and quinolones). Metallic taste

Protective agents:Colloidal bismuth

- Most effects similar to sucralfate: combine with proteins in the ulcer base; forming protective layer over ulcerated, adsorbing pepsin and enhancing the gastric mucosal barrier
- Stimulate PG and mucus production
- Eradicate H. pylori
- Dark discoloration of teeth and tongue

Antacids

- Weak bases, neutralize gastric acid and reduce pepsin activity as it is inactive with PH above 4
- Reduce pain and may promote healing
- Tab act more slowly than liquid antacid unless sucked or chewed

Uses:

- Non-ulcer dyspepsia & mild GERD (given 1 hr & 3 hr after meal & at bed time)

Antacids:Sodium bicarbonate

- Absorbed systematically and should not be used for long-term treatment (alkalosis).
- Has quick onset and short duration
- CO_2 release with development of distension
- Cause alkalinization of urine with precipitation of phosphate stones and decrease excretion of weak bases drugs
- Contraindicated in hypertension or heart failure due to its high sodium content

Antacids: Calcium carbonate

- Partially absorbed, has some systemic effect
- Has quick onset and long duration
- CO_2 release with development of distension
- Can cause hypercalcemia
- Should not be used for long-term treatment
- May stimulate gastrin release causing rebound acid production
- Contraindicated in renal disease due to danger of hypercalcemia and hypercalcuria
- Cause constipation

Antacids (the good ones)

Magnesium trisilicate:

- Not absorbed, no systemic effects or CO_2 release
- Delayed onset with long duration
- Very effective , acts chemically and physically

Mg tricilicate+ Hcl ⟹ Mg cl_2 +H_2O+ silicon dioxide gel

- Causes diarrhea

Antacids (the good ones)

Magnesium hydroxide:

- Not absorbed, no systemic effects or CO_2 release
- Delayed onset with long duration
- Causes diarrhea

Aluminum hydroxide:

- Not absorbed, no systemic effects
- Causes constipation

Combination is logical to produce a balance between the agents' adverse effects on the bowel

Antacids:Adverse effects

- Bind drugs: tetracycline, digoxin and prevent their absorption
- Change in bowel habits
- Calcium-based antacids cause rebound acid secretion
- Decrease the absorption of acidic drugs and enhancing the basic drugs
- Enhance excretion of acidic drugs and inhibit the basic ones

Helicobacter pylori

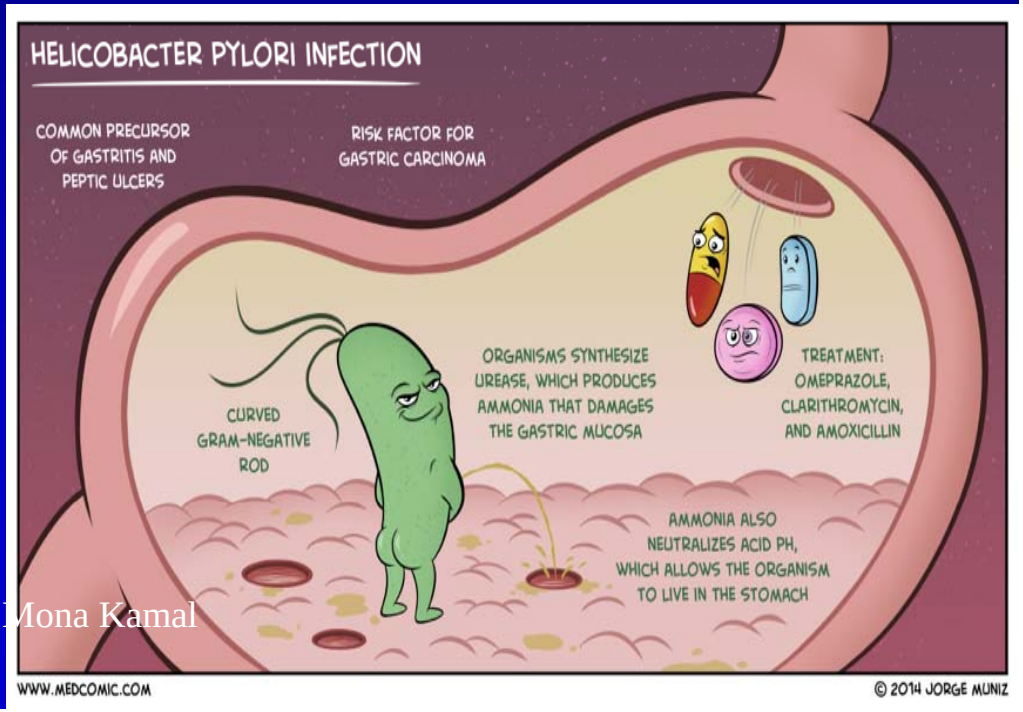
- Most common infection in the world (20%)
- Treated for two weeks by antimicrobials:

1. Metronidazole
2. Amoxycillin
3. Clarithromycin
4. Doxycycline

+ Acid suppressants

+ Colloidal bismuth

*** No antacids



Helicobacter pylori

- Triple therapy

  A proton-pump inhibitor (omeprazole or esomeprazole),

  2 antibiotics (amoxycillin , clarithromycin , doxycycline & metronidazole) for 2 weeks.

- Quadruple therapy (When triple therapy not effective).

  Bismuth subsalicylate was added , a proton pump inhibitor and 2 antibiotics for 2 weeks.

- Provide 90% or more eradication of H. pylori & promote healing of peptic ulcer
- **Note:** PPIs do not kill H. pylori. Promote ulcer healing .

Thanks For Your Attention

Prof. Mona Kamal